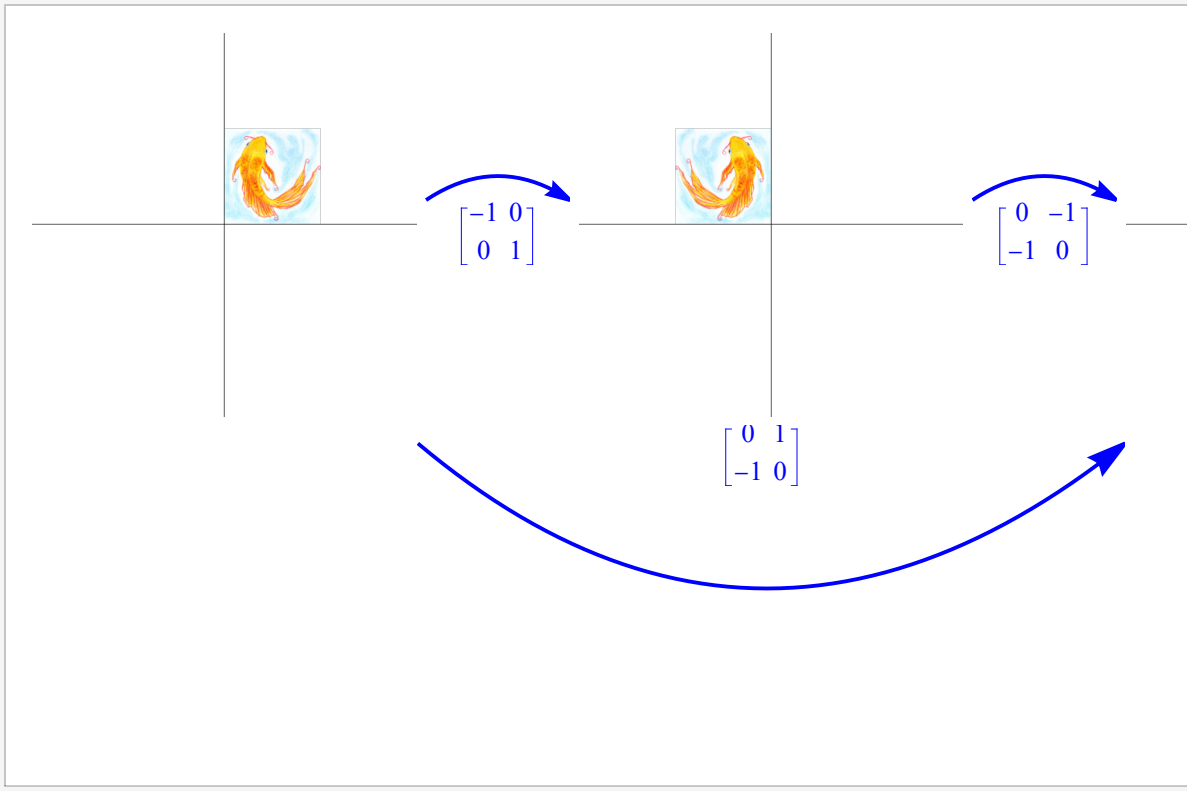
MATH 420-01

2.3 Composition of Linear Transformations and Matrix Multiplication

Out[ ]=



Remark:

Matrix multiplication is defined to make composition of functions (transformations) work.

Example:

Let $T : \mathbb{P}_3 \to M_{2\,x2}$ and $S : M_{2\,x2} \to \mathbb{R}^3$ so $ST : \mathbb{P}_3 \to \mathbb{R}^3$

$$T(f) = \begin{pmatrix} f(1) & f'(0) \\ 0 & f(1) + f''(1) \end{pmatrix}$$

$$S\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \begin{pmatrix} a \\ c \\ b + d \end{pmatrix}$$

Let $\beta = \{1, x, x^2, x^3\}$, $\gamma = \{E^{ij} \mid i, j \in \{1, 2\}\}$ and $\eta = \{e_i \mid i \in \{1, 2, 3\}\}$ be bases for the respective vector spaces above.

E^{ij} is the matrix with a 1 in row i and column j and zeros elsewhere.

e_i is the vector with a 1 in row i and zeros elsewhere.

To find $[T]_\beta^\gamma$, we should get a 4x4 matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 7 \end{bmatrix}$

$$T(1) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T(x) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = 1\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 1\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$T(x^2) = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

$$T(x^3) = \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix}$$

$$[S]_\gamma^\eta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}; \quad S\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \quad S\left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

To find $[ST]_\beta^\eta$, we should get a 3×4 matrix $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 7 \end{bmatrix}$

$$ST(1) = S\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$ST(x) = S\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$ST(x^2) = S\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

$$ST\left(x^3\right) = S\begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}$$

$[S]_\gamma^\eta [T]_\beta^\gamma =$

In[ ]:= $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} . \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 7 \end{pmatrix}$

Out[ ]= $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 7 \end{pmatrix}$

An Aside to Look at Matrix Multiplication in More Detail

Definition:

Let A be an $m \times n$ matrix, let x be an $n \times 1$ vector, and let B be an $n \times p$ matrix, all with entries from the field $\mathbb{F}$. We define

$$Ax = \sum_{k=1}^{n} x_k\, a_k, \text{ where } x_k \text{ is the } k-\text{th entry in } x \text{ and } a_k \text{ is the } k-\text{th column of } A.$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix} = \begin{pmatrix} 10 \\ 28 \\ 46 \end{pmatrix}$$

We define

$AB = \left[Ab_1 \;\; Ab_2 \;\; \ldots Ab_p\right]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 2 & 1 \\ 1 & 1 \end{pmatrix} = \left(\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 10 & 4 \\ 28 & 7 \\ 46 & 10 \end{pmatrix}$$

Observation:

The columns of AB consist of linear combinations of the columns of A using the weights from each column from B.

FUNCTION NOTATION: $g\,(f\,(x))$ means apply f to x, then apply g to $f\,(x)$

Theorem:

Let $T : U \to V$ and $S : V \to W$ be linear transformations on finite dimensional vector spaces, U, V, and W, over the same field $\mathbb{F}$. Let α, β, and γ be bases of each vector space respectively.

Set $B = [T]_\alpha^\beta$ and $A = [S]_\beta^\gamma$. Then $[ST]_\alpha^\gamma = AB$.

Proof:

Let $\{u_1, \ldots, u_q\}$, $\{v_1, \ldots v_r\}$, and $\{w_1, \ldots, w_s\}$ be basis for U, V, and W respectively.